

Codveda Technology-Level 1 Task 1

Data Cleaning and preprocessing

Project Objective

The objective of this project is to clean and preprocess the provided sentiment dataset by identifying and handling missing values, removing duplicates, standardizing inconsistent data formats, and ensuring that variables have appropriate data types for further analysis.

Dataset Overview

The original dataset contains 732 rows and 15 columns. The dataset contains information about social media posts, including sentiment, timestamp, platform, hashtags, engagement metrics and country.

Cleaning Process

The following preprocessing steps were performed:

1. Inspected the structure and data types of the dataset.
2. Checked for missing values.
3. Identified and removed unnecessary index columns.
4. Identified and removed duplicate records.
5. Removed leading and trailing whitespace from text-based columns.
6. Converted the `Timestamp` column to datetime format.
7. Validated the cleaned dataset.

Final Result

After preprocessing, the dataset contained 710 rows and 13 columns, with no missing values or duplicate records.

1. Import Libraries

```
In [2]: import pandas as pd
```

```
In [5]: import os  
  
os.listdir("Downloads")
```

```
Out[5]: ['1769271568315.png',
'1895 The Power Of Silence by Horatio W Dresser - Copy.pdf',
'3) Sentiment dataset.csv',
'Anaconda3-2025.12-2-Windows-x86_64.exe',
'Build_the_Life_You_Want_The_Art_and_Science_of_Getting_Happier.pdf',
'ChatGPT Installer.exe',
'Code_Generated_Image.png',
'desktop.ini',
'DM Vasudevan - Textbook of Biochemistry For Medical Students, 6th Edition (2).pdf',
'download (1).jfif',
'download (2).jfif',
'download (3).jfif',
'download (4).jfif',
'download (5).jfif',
'download.jfif',
'Find_Your_Why_A_Practical_Guide_for_Discovering_Purpose_for_You_.pdf',
'FocusWriter_1.8.13.exe',
'fortnite aura fortnitethumbnails image by @sssnipergamer.jfif',
'GitHub - martinamx_Parks_and_Rec_Create_db_ This project is designed to create and manage a database for a Parks and Recreation department. It includes SQL scripts for database creation, table creation, data insertion, and analytics.html',
'Glycemic_index_and_glycemic_load_Challenges_and_be.pdf',
'LinkedIn_Banner (1).png',
'LinkedIn_Banner (2).png',
'LinkedIn_Banner.png',
'LinkedIn_Networking_Playbook.docx',
'Manchester is Blue.jfif',
'Motivational Gradient Wallpaper - _Nothing Will Work Unless You Do_ (1).jfif',
'Motivational Gradient Wallpaper - _Nothing Will Work Unless You Do_ (2).jfif',
'Motivational Gradient Wallpaper - _Nothing Will Work Unless You Do_.jfif',
'MySQL Beginner series',
'mysql-installer-community-8.0.44.0.msi',
'Overcoming Social Anxiety and Shyness @variableTribe.pdf',
'Parks_and_Rec_Create_db-main.zip',
'parks_and_Rec_Create_db.html',
'PBIDesktopSetup_x64.exe',
'Procrastination.pdf',
'python-3.13.5-amd64.exe',
'TableauPublicDesktop-64bit-2026-1-1.exe',
'The psychology of money .pdf',
'think-again-the-power-of-knowing-what-you-dont-know-random-house-large-print-grant-adam-z-lib.org_.pdf',
'TRUST THE PROCESS.jfif',
```

```
'VSCodeUserSetup-x64-1.101.1.exe',  
'wiztree_4_31_portable',  
'wiztree_4_31_portable.zip']
```

2. Load The Dataset

The sentiment dataset provided for the internship was loaded into a pandas DataFrame for inspection and preprocessing.

```
In [6]: df = pd.read_csv("Downloads/3) Sentiment dataset.csv")
```

3. Initial Data Inspection

The first five rows

```
In [7]: df.head()
```

Out[7]:

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023
2	2	2	Just finished an amazing workout! 💪 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023
3	3	3	Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023
4	4	4	Trying out a new recipe for dinner tonight. ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food	12.0	25.0	Australia	2023



In [8]: df.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 732 entries, 0 to 731
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0.1    732 non-null   int64
1   Unnamed: 0      732 non-null   int64
2   Text            732 non-null   object
3   Sentiment       732 non-null   object
4   Timestamp       732 non-null   object
5   User            732 non-null   object
6   Platform        732 non-null   object
7   Hashtags        732 non-null   object
8   Retweets        732 non-null   float64
9   Likes           732 non-null   float64
10  Country         732 non-null   object
11  Year            732 non-null   int64
12  Month           732 non-null   int64
13  Day             732 non-null   int64
14  Hour            732 non-null   int64
dtypes: float64(2), int64(6), object(7)
memory usage: 85.9+ KB

```

4. Duplicate Records

Duplicate records were checked using pandas' `duplicated()` method.

The initial checked was performed before removing the unnecessary index columns. After those columns were removed, 22 genuine duplicate records were identified and subsequently removed.

```
In [10]: df.duplicated().sum()
```

```
Out[10]: np.int64(0)
```

```
In [11]: df[["Unnamed: 0", "Unnamed: 0.1"]].head(10)
```

Out[11]:

	Unnamed: 0	Unnamed: 0.1
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

In [12]: `df["Unnamed: 0"].equals(df["Unnamed: 0.1"])`

Out[12]: False

In [13]: `df[["Unnamed: 0", "Unnamed: 0.1"]].tail(10)`

Out[13]:

	Unnamed: 0	Unnamed: 0.1
722	727	723
723	728	724
724	729	725
725	730	726
726	731	727
727	732	728
728	733	729
729	734	730
730	735	731
731	736	732

In [14]: `df["Unnamed: 0"].is_unique`

Out[14]: True

In [15]: `df["Unnamed: 0.1"].is_unique`

Out[15]: True

In [16]: `df[["Unnamed: 0", "Unnamed: 0.1"]].describe()`

Out[16]:

	Unnamed: 0	Unnamed: 0.1
count	732.000000	732.000000
mean	369.740437	366.464481
std	212.428936	211.513936
min	0.000000	0.000000
25%	185.750000	183.750000
50%	370.500000	366.500000
75%	553.250000	549.250000
max	736.000000	732.000000

In [17]: `df[["Unnamed: 0", "Unnamed: 0.1"]].tail(10)`

Out[17]:

	Unnamed: 0	Unnamed: 0.1
722	727	723
723	728	724
724	729	725
725	730	726
726	731	727
727	732	728
728	733	729
729	734	730
730	735	731
731	736	732

In [18]: `df[df["Unnamed: 0"] != df["Unnamed: 0.1"]][["Unnamed: 0", "Unnamed: 0.1"]].head(20)`

Out[18]:

	Unnamed: 0	Unnamed: 0.1
26	28	27
27	29	28
28	30	29
29	31	30
30	32	31
31	33	32
32	34	33
33	35	34
34	36	35
35	37	36
36	38	37
37	39	38
38	40	39
39	41	40
40	42	41
41	43	42
42	44	43
43	45	44
44	46	45
45	47	46

5. Removing unnecessary columns

The dataset contained two columns `Unnamed: 0` and `Unnamed: 0.1`, which were identified as redundant index-tracking columns rather than meaningful analytical variables.

Both columns were removed to retain only variables relevant to the analysis.

```
In [19]: df = df.drop(columns=["Unnamed: 0", "Unnamed: 0.1"])
```

Verification

The column names were checked after the removal to confirm that both unnecessary columns have been successfully dropped.

```
In [20]: df.columns
```

```
Out[20]: Index(['Text', 'Sentiment', 'Timestamp', 'User', 'Platform', 'Hashtags',  
              'Retweets', 'Likes', 'Country', 'Year', 'Month', 'Day', 'Hour'],  
             dtype='object')
```

```
In [21]: df.dtypes
```

```
Out[21]: Text          object  
Sentiment          object  
Timestamp          object  
User              object  
Platform          object  
Hashtags          object  
Retweets          float64  
Likes             float64  
Country           object  
Year              int64  
Month             int64  
Day               int64  
Hour              int64  
dtype: object
```

```
In [23]: df["Timestamp"].head(10)
```

```
Out[23]: 0    2023-01-15 12:30:00
         1    2023-01-15 08:45:00
         2    2023-01-15 15:45:00
         3    2023-01-15 18:20:00
         4    2023-01-15 19:55:00
         5    2023-01-16 09:10:00
         6    2023-01-16 14:45:00
         7    2023-01-16 19:30:00
         8    2023-01-17 08:00:00
         9    2023-01-17 12:20:00
         Name: Timestamp, dtype: object
```

```
In [25]: pd.to_datetime(df["Timestamp"], errors="coerce").isna().sum()
```

```
Out[25]: np.int64(0)
```

Data type standardization

The `Timestamp` column was initially stored as an object(text) data type. Since it represents date and time information, it was converted to pandas' datetime format.

The `Retweeta` and `Likes` columns were also converted from floating-point values to integers because they contain whole number engagement counts.

```
In [26]: df["Timestamp"] = pd.to_datetime(df["Timestamp"], errors="coerce")
```

```
In [27]: df["Timestamp"].dtype
```

```
Out[27]: dtype('<M8[ns]')
```

```
In [28]: df["Sentiment"].unique()
```

```

Out[28]: array([' Positive ', ' Negative ', ' Neutral ', ' Anger ',
' Fear ', ' Sadness ', ' Disgust ',
' Happiness ', ' Joy ', ' Love ',
' Amusement ', ' Enjoyment ', ' Admiration ',
' Affection ', ' Awe ', ' Disappointed ',
' Surprise ', ' Acceptance ', ' Adoration ',
' Anticipation ', ' Bitter ', ' Calmness ',
' Confusion ', ' Excitement ', ' Kind ',
' Pride ', ' Shame ', ' Confusion ', ' Excitement ',
' Shame ', ' Elation ', ' Euphoria ', ' Contentment ',
' Serenity ', ' Gratitude ', ' Hope ',
' Empowerment ', ' Compassion ', ' Tenderness ',
' Arousal ', ' Enthusiasm ', ' Fulfillment ',
' Reverence ', ' Compassion ', ' Fulfillment ', ' Reverence ',
' Elation ', ' Despair ', ' Grief ',
' Loneliness ', ' Jealousy ', ' Resentment ',
' Frustration ', ' Boredom ', ' Anxiety ',
' Intimidation ', ' Helplessness ', ' Envy ',
' Regret ', ' Disgust ', ' Despair ',
' Loneliness ', ' Frustration ', ' Anxiety ', ' Intimidation ',
' Helplessness ', ' Jealousy ', ' Curiosity ',
' Indifference ', ' Confusion ', ' Numbness ',
' Melancholy ', ' Nostalgia ', ' Ambivalence ',
' Acceptance ', ' Determination ', ' Serenity ',
' Numbness ', ' Zest ', ' Contentment ', ' Hopeful ', ' Proud ',
' Grateful ', ' Empathetic ', ' Compassionate ', ' Playful ',
' Free-spirited ', ' Inspired ', ' Confident ', ' Serenity ',
' Curiosity ', ' Ambivalence ', ' Despair ', ' Bitterness ',
' Yearning ', ' Fearful ', ' Apprehensive ', ' Overwhelmed ',
' Jealous ', ' Devastated ', ' Frustrated ', ' Envious ',
' Dismissive ', ' Awe ', ' Determination ',
' Nostalgia ', ' Thrill ', ' Calmness ',
' Overwhelmed ', ' Gratitude ', ' Bittersweet ',
' Curiosity ', ' Admiration ', ' Overjoyed ',
' Inspiration ', ' Motivation ', ' Amusement ',
' Contemplation ', ' JoyfulReunion ', ' Excitement ',
' Satisfaction ', ' Blessed ', ' Anticipation ',
' Reflection ', ' Nostalgia ', ' Appreciation ',
' Confidence ', ' Surprise ', ' Accomplishment ',
' Wonderment ', ' Optimism ', ' Pride ',
' Happiness ', ' Curiosity ', ' Enchantment ',
' Intrigue ', ' PlayfulJoy ', ' Mindfulness ',

```

```
' DreamChaser ', ' Elegance ', ' Whimsy ', ' Pensive ',
' Thrill ', ' Harmony ', ' Creativity ', ' Radiance ',
' Wonder ', ' Rejuvenation ', ' Inspiration ',
' Coziness ', ' Gratitude ', ' Adventure ', ' Euphoria ',
' Awe ', ' Melodic ', ' FestiveJoy ',
' InnerJourney ', ' Freedom ', ' Dazzle ',
' Adrenaline ', ' Harmony ', ' ArtisticBurst ',
' Radiance ', ' Wonder ', ' Inspiration ',
' CulinaryOdyssey ', ' Euphoria ', ' Curiosity ',
' Resilience ', ' Immersion ', ' Nostalgia ', ' Spark ',
' Gratitude ', ' Marvel ', ' Serenity ',
' Heartbreak ', ' Loneliness ', ' Grief ',
' Despair ', ' Betrayal ', ' Suffering ',
' EmotionalStorm ', ' Regret ', ' Isolation ',
' Disappointment ', ' LostLove ', ' Melancholy ', ' Exhaustion ',
' Sorrow ', ' Darkness ', ' Desperation ', ' Ruins ',
' Desolation ', ' Regret ', ' Grief ', ' Heartbreak ',
' Betrayal ', ' Resilience ', ' Sorrow ', ' Loss ', ' Heartache ',
' Solitude ', ' Joy ', ' Happiness ', ' Enthusiasm ',
' Gratitude ', ' Positivity ', ' Kindness ', ' Friendship ',
' Love ', ' Surprise ', ' Success ', ' Thrill ', ' Reflection ',
' Enchantment ', ' Exploration ', ' Awe ', ' Amazement ',
' Romance ', ' Captivation ', ' Wonder ', ' Tranquility ',
' Grandeur ', ' Emotion ', ' Energy ', ' Celebration ', ' Charm ',
' Ecstasy ', ' Hope ', ' Creativity ', ' Colorful ', ' Pride ',
' Hypnotic ', ' Connection ', ' Iconic ', ' Euphoria ',
' Journey ', ' Engagement ', ' Touched ', ' Suspense ',
' Satisfaction ', ' Admiration ', ' Triumph ', ' Heartwarming ',
' Obstacle ', ' Sympathy ', ' Pressure ', ' Renewed Effort ',
' Miscalculation ', ' Challenge ', ' Solace ', ' Breakthrough ',
' Harmony ', ' Joy in Baking ', ' Envisioning History ',
' Imagination ', ' Vibrancy ', ' Mesmerizing ',
' Culinary Adventure ', ' Winter Magic ', ' Thrilling Journey ',
" Nature's Beauty ", ' Celestial Wonder ',
' Creative Inspiration ', ' Runway Creativity ',
" Ocean's Freedom ", ' Whispers of the Past ', ' Boredom ',
' Indifference ', ' Disgust ', ' Relief ', ' Positive ',
' Embarrassed ', ' Mischievous ', ' Sad ', ' Hate ', ' Bad ',
' Neutral ', ' Happy '], dtype=object)
```

```
In [29]: df["Sentiment"].unique()
```

Out[29]: 279

In [30]: `df["Sentiment"].value_counts()`

Out[30]:

Sentiment	
Positive	44
Joy	42
Excitement	32
Neutral	14
Contentment	14
..	
Adrenaline	1
Harmony	1
ArtisticBurst	1
Radiance	1
Elegance	1

Name: count, Length: 279, dtype: int64

In [32]: `df["Sentiment"].str.strip().nunique()`

Out[32]: 191

In [33]: `df["Sentiment"].str.strip().value_counts().head(20)`

```
Out[33]: Sentiment
         Positive    45
         Joy        44
         Excitement  37
         Contentment 19
         Neutral    18
         Gratitude  18
         Curiosity  16
         Serenity   15
         Happy      14
         Nostalgia  11
         Despair    11
         Grief       9
         Awe         9
         Sad         9
         Hopeful     9
         Loneliness  9
         Embarrassed 8
         Acceptance  8
         Confusion   8
         Euphoria    7
         Name: count, dtype: int64
```

Text Standardization

Leading and trailing whitespace was removed from the relevant text-based columns to standardize categorical and textual values.

After cleaning, the number of unique values in the `Sentiment` column decreased from 279 to 191, indicating that whitespace inconsistencies had caused multiple representations of the otherwise identical values.

```
In [38]: text_columns = ["Text", "Sentiment", "User", "Platform", "Hashtags", "Country"]

for column in text_columns:
    df[column] = df[column].str.strip()
```

```
In [39]: df["Sentiment"].nunique()
```

```
Out[39]: 191
```



```
In [40]: df["Sentiment"].value_counts().head(20)
```

```
Out[40]: Sentiment
Positive      45
Joy           44
Excitement    37
Contentment   19
Neutral       18
Gratitude     18
Curiosity    16
Serenity      15
Happy         14
Nostalgia     11
Despair       11
Grief         9
Awe           9
Sad           9
Hopeful       9
Loneliness    9
Embarrassed   8
Acceptance    8
Confusion     8
Euphoria      7
Name: count, dtype: int64
```

```
In [41]: df["Platform"].nunique()
```

```
Out[41]: 3
```

```
In [42]: df["Country"].nunique()
```

```
Out[42]: 33
```

```
In [43]: df["Platform"].value_counts()
```

```
Out[43]: Platform
Instagram    258
Twitter      243
Facebook     231
Name: count, dtype: int64
```

```
In [44]: df["Country"].value_counts()
```

```
Out[44]: Country
USA          188
UK           143
Canada       135
Australia    75
India        70
Brazil       17
France       16
Japan        15
Germany      14
Italy        11
Spain        6
South Africa 6
Greece       5
Netherlands  4
Switzerland  3
Portugal     2
Austria      2
Belgium      2
Denmark      2
Czech Republic 2
Sweden       2
Colombia     1
Scotland     1
Kenya        1
Jamaica      1
Ireland      1
China        1
Norway       1
Cambodia     1
Maldives     1
Peru         1
Jordan       1
Thailand     1
Name: count, dtype: int64
```

```
In [45]: df[["Retweets", "Likes"]].describe()
```

Out[45]:

	Retweets	Likes
count	732.000000	732.000000
mean	21.508197	42.901639
std	7.061286	14.089848
min	5.000000	10.000000
25%	17.750000	34.750000
50%	22.000000	43.000000
75%	25.000000	50.000000
max	40.000000	80.000000

In [46]: `df[["Retweets", "Likes"]].apply(lambda x: (x % 1 != 0).sum())`

Out[46]: Retweets 0
 Likes 0
 dtype: int64

In [48]: `df["Retweets"] = df["Retweets"].astype(int)`
`df["Likes"] = df["Likes"].astype(int)`

In [49]: `df[["Retweets", "Likes"]].dtypes`

Out[49]: Retweets int64
 Likes int64
 dtype: object

6. Missing Values Check

Missing values were checked across all columns using the `isnull().sum()` method.

the initial inspection showed that all columns contained 732 non-null values, indicating that there were no missing values in the original dataset.

```
In [50]: df.isnull().sum()
```

```
Out[50]: Text          0  
Sentiment            0  
Timestamp            0  
User                 0  
Platform             0  
Hashtags             0  
Retweets             0  
Likes               0  
Country              0  
Year                 0  
Month                0  
Day                  0  
Hour                 0  
dtype: int64
```

```
In [51]: df.duplicated().sum()
```

```
Out[51]: np.int64(22)
```

```
In [52]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 732 entries, 0 to 731
Data columns (total 13 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Text        732 non-null    object
 1   Sentiment   732 non-null    object
 2   Timestamp   732 non-null    datetime64[ns]
 3   User        732 non-null    object
 4   Platform    732 non-null    object
 5   Hashtags    732 non-null    object
 6   Retweets    732 non-null    int64
 7   Likes       732 non-null    int64
 8   Country     732 non-null    object
 9   Year        732 non-null    int64
10  Month       732 non-null    int64
11  Day         732 non-null    int64
12  Hour        732 non-null    int64
dtypes: datetime64[ns](1), int64(6), object(6)
memory usage: 74.5+ KB
```

```
In [53]: df[df.duplicated(keep=False)].sort_values("Text").head(10)
```

Out[53]:

	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year
292	A compassionate rain, tears of empathy falling...	Compassionate	2021-07-01 12:10:00	RainNurturer	Instagram	#Compassionate #TearsOfEmpathy	21	42	Canada	2021
278	A compassionate rain, tears of empathy falling...	Compassionate	2021-07-01 12:10:00	RainNurturer	Instagram	#Compassionate #TearsOfEmpathy	21	42	Canada	2021
285	A free spirit soaring on the wings of dreams, ...	Free-spirited	2020-06-10 10:05:00	DreamSoarer	Twitter	#FreeSpirit #WingsOfDreams	22	44	UK	2020
271	A free spirit soaring on the wings of dreams, ...	Free-spirited	2020-06-10 10:05:00	DreamSoarer	Twitter	#FreeSpirit #WingsOfDreams	22	44	UK	2020
267	A playful escapade in the carnival of life, ca...	Playful	2018-08-22 17:20:00	CarnivalDreamer	Facebook	#Playful #CarnivalEscapade	24	48	Australia	2018
281	A playful escapade in the carnival of life, ca...	Playful	2018-08-22 17:20:00	CarnivalDreamer	Facebook	#Playful #CarnivalEscapade	24	48	Australia	2018
295	A playful escapade in the carnival of life, ca...	Playful	2018-08-22 17:20:00	CarnivalDreamer	Facebook	#Playful #CarnivalEscapade	24	48	Australia	2018
272	Bathed in the golden hues of	Grateful	2022-04-01 18:30:00	SunsetAdmirer	Instagram	#Grateful #GoldenHues	19	38	Australia	2022

	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year
	gratefulness, a S...									
286	Bathed in the golden hues of gratefulness, a S...	Grateful	2022-04-01 18:30:00	SunsetAdmirer	Instagram	#Grateful #GoldenHues	19	38	Australia	2022
291	Charting a course through the waves of hopeful...	Hopeful	2022-10-30 11:40:00	WaveNavigator	Twitter	#Hopeful #SailorOfDreams	17	34	Australia	2022

Removing Duplicate Records

After the unnecessary index columns were removed, 22 duplicate records were identified which were later confirmed and removed using `drop_duplicates()`.

```
In [54]: df = df.drop_duplicates()
```

Duplicate Validation

The dataset was checked again after removing the duplicates to confirm that no duplicate records remained.

```
In [55]: df.duplicated().sum()
```

```
Out[55]: np.int64(0)
```

```
In [56]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 710 entries, 0 to 731
Data columns (total 13 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Text        710 non-null    object
 1   Sentiment   710 non-null    object
 2   Timestamp   710 non-null    datetime64[ns]
 3   User        710 non-null    object
 4   Platform    710 non-null    object
 5   Hashtags    710 non-null    object
 6   Retweets    710 non-null    int64
 7   Likes       710 non-null    int64
 8   Country     710 non-null    object
 9   Year        710 non-null    int64
10  Month       710 non-null    int64
11  Day         710 non-null    int64
12  Hour        710 non-null    int64
dtypes: datetime64[ns](1), int64(6), object(6)
memory usage: 77.7+ KB
```

Export Cleaned Dataset

The cleaned dataset was exported as CSV file for use in subsequent exploratory data analysis and other analytical tasks.

The original DataFrame index was excluded from the exported file to prevent the creation of an unnecessary index column.

```
In [58]: df.to_csv("cleaned_sentiment_dataset.csv", index=False)
```

```
In [59]: import os

os.getcwd()
```

```
Out[59]: 'C:\\Users\\HP'
```

```
In [60]: os.listdir()
```



```
Out[60]: ['.anaconda',  
          '.conda',  
          '.continuum',  
          '.idlerc',  
          '.ipynb_checkpoints',  
          '.ipython',  
          '.jupyter',  
          '.matplotlib',  
          '.python_history',  
          '.vscode',  
          '3D Objects',  
          'anaconda3',  
          'AppData',  
          'Application Data',  
          'battery-report.html',  
          'cleaned_sentiment_dataset.csv',  
          'Codveda_Level_1_Task_1_Data_Cleaning.ipynb',  
          'Codveda_Level_1_Task_2_EDA.ipynb',  
          'Codveda_Level_1_Task_2_EDA.py',  
          'Contacts',  
          'Cookies',  
          'Desktop',  
          'DOC-20250519-WA0011..pptx',  
          'Documents',  
          'Downloads',  
          'Favorites',  
          'learning.py',  
          'Links',  
          'Local Settings',  
          'MicrosoftEdgeBackups',  
          'Music',  
          'My Documents',  
          'NetHood',  
          'NTUSER.DAT',  
          'ntuser.dat.LOG1',  
          'ntuser.dat.LOG2',  
          'NTUSER.DAT{b12ad8bb-dcde-11ef-b0d8-e86f3869a1f5}.TM.blf',  
          'NTUSER.DAT{b12ad8bb-dcde-11ef-b0d8-e86f3869a1f5}.TMContainer000000000000000001.regtrans-ms',  
          'NTUSER.DAT{b12ad8bb-dcde-11ef-b0d8-e86f3869a1f5}.TMContainer000000000000000002.regtrans-ms',  
          'ntuser.ini',  
          'OneDrive',  
          'Pictures',
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'PrintHood',  
'Recent',  
'Saved Games',  
'Searches',  
'SEMINAR SLIDE Masabaa.pptx',  
'SendTo',  
'something.py',  
'Start Menu',  
'Templates',  
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'Videos',  
'waleed']
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